CSE 464 Project Part 3 README

Student Info
Name: Vamika Negi
ASU ID: 1226812251
Repository: https://github.com/vamika27/CSE4642026-vnegi3
Branch: refactor
Pull Request: https://github.com/vamika27/CSE4642026-vnegi3/pull/1

Project Overview
This project refactors the graph search system using software design patterns. It includes five refactoring steps, implementation of Template and Strategy patterns, and a new Random Walk algorithm with dead-end handling.

Build Instructions
mvn clean package

Run Instructions
java -cp target/CSE4642026-vnegi3-1.0-SNAPSHOT.jar graph.Main

Refactoring Commits

Refactor 1: Format & Clean Project
https://github.com/vamika27/CSE4642026-vnegi3/commit/ea1c14c

Refactor 2: Extract Node Validation
https://github.com/vamika27/CSE4642026-vnegi3/commit/8687b42

Refactor 3: Extract Path Reconstruction
https://github.com/vamika27/CSE4642026-vnegi3/commit/6843adc

Refactor 4: Extract Neighbor Lookup
https://github.com/vamika27/CSE4642026-vnegi3/commit/322d27c

Refactor 5: Extract Frontier Selection

https://github.com/vamika27/CSE4642026-vnegi3/commit/76743be

Template Pattern
Template Pattern implemented via GraphSearchTemplate, which defines the common structure of graph traversal including validation, frontier handling, expansion, and path building.

Commit:
https://github.com/vamika27/CSE4642026-vnegi3/commit/4b0d7ef

Strategy Pattern
Strategy Pattern implemented using:
- SearchStrategy interface
- BFSStrategy, DFSStrategy, RandomWalkStrategy
- SearchStrategyFactory for runtime selection

Commit:
https://github.com/vamika27/CSE4642026-vnegi3/commit/76d87d3

Random Walk Search
Random Walk algorithm added with:
- Random neighbor traversal
- Dead-end detection
- Restart mechanism to ensure valid path discovery

Initial Implementation:
https://github.com/vamika27/CSE4642026-vnegi3/commit/2dba610

Fix (Dead-End Restart):
https://github.com/vamika27/CSE4642026-vnegi3/commit/b229947

Example Execution
Input file:
sample_inputs/input1.dot

Example outputs:
BFS: a -> e -> f -> h

DFS: a -> e -> g -> h
Random Walk: performs multiple attempts and restarts when a dead end is encountered until a valid path is found

Required Screenshots

- Build success output (mvn package)

```
vamikanegi@Vamikas-MacBook-Air CSE4642026-vnegi3 % mvn clean package
[INFO] Scanning for projects...
[INFO]
[INFO] -------------------< asu.cse464:CSE4642026-vnegi3 >--------------------
[INFO] Building CSE 464 Course Project 1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- clean:3.2.0:clean (default-clean) @ CSE4642026-vnegi3 ---
[INFO] Deleting /Users/vamikanegi/CSE464/CSE4642026-vnegi3/target
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ CSE4642026-vnegi3 ---
[INFO] Copying 0 resource from src/main/resources to target/classes
[INFO]
[INFO] --- compiler:3.15.0:compile (default-compile) @ CSE4642026-vnegi3 ---
[INFO] Recompiling the module because of changed source code.
[INFO] Compiling 14 source files with javac [debug target 17] to target/classes
[WARNING] location of system modules is not set in conjunction with -source 17
  not setting the location of system modules may lead to class files that cannot run on JDK 17
    --release 17 is recommended instead of -source 17 -target 17 because it sets the location of system modules automatically
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @ CSE4642026-vnegi3 ---
[INFO] skip non existing resourceDirectory /Users/vamikanegi/CSE464/CSE4642026-vnegi3/src/test/resources
[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ CSE4642026-vnegi3 ---
[INFO] Recompiling the module because of changed dependency.
[INFO] Compiling 1 source file with javac [debug target 17] to target/test-classes
[WARNING] location of system modules is not set in conjunction with -source 17
  not setting the location of system modules may lead to class files that cannot run on JDK 17
    --release 17 is recommended instead of -source 17 -target 17 because it sets the location of system modules automatically
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ CSE4642026-vnegi3 ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running graph.GraphTest
[INFO] Tests run: 22, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.209 s -- in graph.GraphTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 22, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- jar:3.5.0:jar (default-jar) @ CSE4642026-vnegi3 ---
[INFO] Building jar: /Users/vamikanegi/CSE464/CSE4642026-vnegi3/target/CSE4642026-vnegi3-1.0-SNAPSHOT.jar
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  1.487 s
[INFO] Finished at: 2026-05-03T19:43:08-07:00
[INFO] ------------------------------------------------------------------------
```

- BFS execution output

```
vamikanegi@Vamikas-MacBook-Air CSE4642026-vnegi3 % java -cp target/CSE4642026-vnegi3-1.0-SNAPSHOT.jar graph.Main
===== BFS =====
a -> e -> f -> h

===== DFS =====
a -> e -> g -> h

===== RANDOM WALK RUN 1 =====
random testing
attempt 1
visiting a
visiting a -> e
visiting a -> e -> f
visiting a -> e -> f -> h
dead end, restarting
attempt 2
visiting a
visiting a -> e
visiting a -> e -> f
visiting a -> e -> f -> h
dead end, restarting
attempt 3
visiting a
visiting a -> b
visiting a -> b -> c
a -> b -> c

===== RANDOM WALK RUN 2 =====
random testing
attempt 1
visiting a
visiting a -> b
visiting a -> b -> c
a -> b -> c

===== RANDOM WALK RUN 3 =====
random testing
attempt 1
visiting a
visiting a -> e
visiting a -> e -> g
visiting a -> e -> g -> h
dead end, restarting
attempt 2
visiting a
visiting a -> b
visiting a -> b -> c
a -> b -> c
```

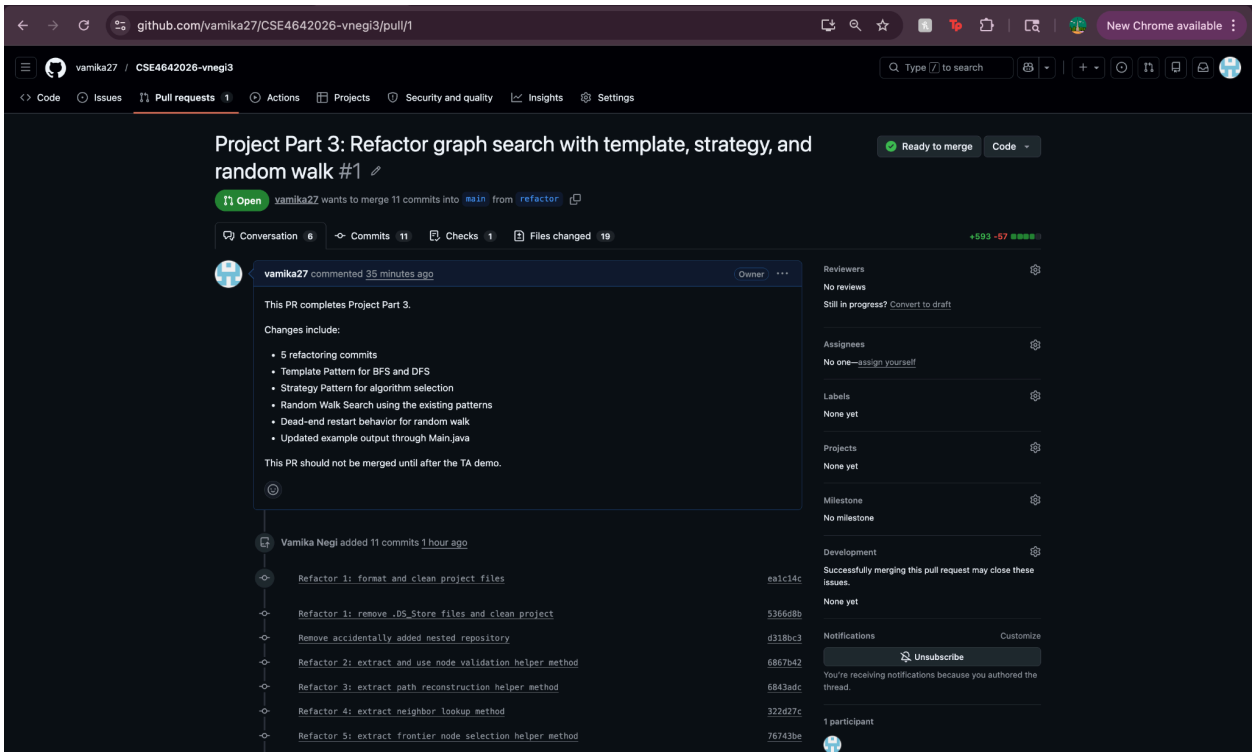
- DFS execution output

```
vamikanegi@Vamikas-MacBook-Air CSE4642026-vnegi3 % java -cp target/CSE4642026-vnegi3-1.0-SNAPSHOT.jar graph.Main
===== BFS =====
a -> e -> f -> h

===== DFS =====
a -> e -> g -> h

===== RANDOM WALK RUN 1 =====
random testing
attempt 1
visiting a
visiting a -> e
visiting a -> e -> f
visiting a -> e -> f -> h
dead end, restarting
attempt 2
visiting a
visiting a -> e
visiting a -> e -> f
visiting a -> e -> f -> h
dead end, restarting
attempt 3
visiting a
visiting a -> b
visiting a -> b -> c
a -> b -> c

===== RANDOM WALK RUN 2 =====
random testing
attempt 1
visiting a
visiting a -> b
visiting a -> b -> c
a -> b -> c

===== RANDOM WALK RUN 3 =====
random testing
attempt 1
visiting a
visiting a -> e
visiting a -> e -> g
visiting a -> e -> g -> h
dead end, restarting
attempt 2
visiting a
visiting a -> b
visiting a -> b -> c
a -> b -> c
```

- Random walk runs showing restart behavior

```
vamikanegi@Vamikas-MacBook-Air CSE4642026-vnegi3 % java -cp target/CSE4642026-vnegi3-1.0-SNAPSHOT.jar graph.Main
===== BFS =====
a -> e -> f -> h

===== DFS =====
a -> e -> g -> h

===== RANDOM WALK RUN 1 =====
random testing
attempt 1
visiting a
visiting a -> e
visiting a -> e -> f
visiting a -> e -> f -> h
dead end, restarting
attempt 2
visiting a
visiting a -> e
visiting a -> e -> f
visiting a -> e -> f -> h
dead end, restarting
attempt 3
visiting a
visiting a -> b
visiting a -> b -> c
a -> b -> c

===== RANDOM WALK RUN 2 =====
random testing
attempt 1
visiting a
visiting a -> b
visiting a -> b -> c
a -> b -> c

===== RANDOM WALK RUN 3 =====
random testing
attempt 1
visiting a
visiting a -> e
visiting a -> e -> g
visiting a -> e -> g -> h
dead end, restarting
attempt 2
visiting a
visiting a -> b
visiting a -> b -> c
a -> b -> c
```

- Pull request page with comments



- PR Comments

Comments

Review | Conversation | Filter comments

Showing 5 of 5 comments

`src/main/java/graph/Graph.java` Line 70

```
70  -   public Path GraphSearch(String src, String dst, Algorithm algo) {
```

vamika27 26 minutes ago — Owner, Author

Refactored graph search to delegate algorithm selection using strategy pattern, improving modularity and extensibility.

No replies

`src/main/java/graph/GraphSearchTemplate.java` Line 7

```
7   +   public Path search(Graph graph, String src, String dst) {
```

vamika27 26 minutes ago — Owner, Author

Template pattern introduced here to define a consistent structure for all graph search algorithms.

No replies

`src/main/java/graph/RandomWalkSearch.java` Line 42

```
42  +                   System.out.println("dead end, restarting");
```

vamika27 25 minutes ago — Owner, Author

Added restart logic to handle dead ends and ensure the random walk eventually finds a valid path if one exists.

No replies

`src/main/java/graph/RandomWalkStrategy.java` Line 8

```
8   +   public Path search(Graph graph, String src, String dst) {
```

vamika27 25 minutes ago — Owner, Author

Strategy pattern implementation allows dynamic selection of graph traversal algorithms at runtime.

No replies

`src/main/java/graph/SearchStrategyFactory.java` Line 5

```
5   +   public static SearchStrategy getStrategy(Algorithm algo) {
```

vamika27 25 minutes ago — Owner, Author

Factory pattern used to encapsulate algorithm selection logic and decouple it from the Graph class.

No replies

Pull Request Review
https://github.com/vamika27/CSE4642026-vnegi3/pull/1

Includes:
- Template pattern explanation
- Strategy pattern explanation
- Factory pattern explanation
- Random walk restart logic
- Refactoring justification